Solar Tracker — Lab Results & Efficiency Calculation

Arduino-Based Dual-Axis Solar Tracking System

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1. Test Conditions

Test Type	Test Date	Measurement Window	Location	Sky Condition
Tracker	2026-09-03	12:30 PM – 1:30 PM (1h peak / solar-noon window)	Niagara, ON, Canada	Cloudy (Partial)
Fixed	2026-09-04	6:41 AM – 7:51 PM (full daylight, continuous)	Niagara, ON, Canada	Cloudy (Partial)

Additional notes:

- Panel rated output: 1.2 W (5V)
- Reference daylight window: Sunrise 6:41 AM | Sunset 7:51 PM | Total daylight 13h 10m
- Two separate calendar days were used for this test: the tracker was sampled at its peak (solar-noon) window on 2026-09-03, and the fixed panel was logged continuously for the full daylight period on 2026-09-04, once continuous wall-powered logging became available.
- For the standing panel test, both Servos will be fixed to 90°.

2. Methodology

Two data sets were collected to compare energy capture between a fixed (static) panel and the Arduino-controlled dual-axis tracker. Because continuous full-day logging could only be dedicated to one configuration at a time, the two panels were tested on separate days rather than simultaneously:

- Fixed panel (2026-09-04): energy logged continuously across the full daylight window (13h 10m) to obtain α, in Wh.
- Tracker (2026-09-03): energy was measured over a 1-hour peak-sun window (12:30 PM – 1:30 PM, centered on solar noon) to obtain β, in Wh, and extrapolated across the full daylight period, assuming 100% tracking performance (panel always oriented toward the sun).

The extrapolation multiplier converts the 1-hour peak-window sample to the full 13h 10m daylight period:

$$\boldsymbol{\gamma = 13h\ 10m \div 1h = 13 + (10/60) = 13.17h}$$

Extrapolated tracker energy:

$$\boldsymbol{x = \beta \times \gamma = \beta \times 13.17\ \ [Wh]}$$

Tracking efficiency gain (energy captured by tracker vs. fixed panel):

$$\boldsymbol{y = (x - \alpha)/\alpha \times 100\ \ [\%]}$$

Note: α is a direct, continuously-measured full-day total, while x — a single 1-hour peak-window sample extrapolated (×13.17) to represent a full day. Because the two configurations were not measured the same way, x and the resulting efficiency gain y should be read as an estimation of tracker performance, not a directly measured full-day value.

3. Results

Parameter	Symbol	Value
Fixed panel energy (measured, 13h 10m, 2026-09-04)	α	13.59 Wh
Tracker energy (measured, 1h peak-window sample, 2026-09-03)	β	1.27Wh
Extrapolation multiplier (13h 10m ÷ 1h)	γ	13.17h
Extrapolated tracker energy (full daylight, estimated)	x	17.40 Wh
Tracking efficiency gain (estimated)	y	28.04%

4. Sources of Error

- Non-simultaneous testing: Tracker and fixed panel were measured on different days (09-03 vs 09-04), so differing sunlight/cloud cover between days confounds the comparison.
- Extrapolation assumption: Tracker's full-day energy was estimated from one 1-hour peak-sun sample × 13.17, assuming 100% tracking accuracy all day — likely overstates real output.
- Single-trial measurements: Each configuration was tested only once, so there's no way to confirm the 28.04% figure is repeatable.
- Sensor/instrumentation error: ACS712 current sensor and Arduino's 9600 baud could compound errors calculations (Temperature, Humidity, Voltage, Current, Power).
- Mechanical tracking error: Servo backlash and photoresistor response lag mean the panel doesn't track the sun with perfect precision, unlike the theoretical assumption.
- Fixed panel angle: Locked at 90° rather than an optimal tilt for the date/latitude, possibly understating a realistic fixed-panel baseline.

5. Summary

Over the recorded daylight period (13h 10m), the fixed panel captured 13.59 Wh on 2026-09-04, while the tracking panel — extrapolated from a 1-hour peak-window sample taken on 2026-09-03 at full tracking performance — is estimated to capture 17.40 Wh, representing an estimated **3.81 Wh (28.04%)** increase in energy capture efficiency compared to the fixed configuration.

These results demonstrate the approximate electrical energy gain (Wh) achievable through active dual-axis solar tracking using low-cost, accessible components (Arduino Nano, servo motors, and photoresistor-based light sensing).